Structural Analysis and Design Report

Project: 23-Story High-Rise Building in Rupnagar, Punjab

1. Project Overview

This project involves the structural modeling and seismic evaluation of a 23-story reinforced concrete structure. The building features significant vertical and horizontal setbacks, requiring detailed analysis of mass-stiffness centers and lateral load distribution. The project was modeled and analyzed using ETABS Ultimate v22.7.0.

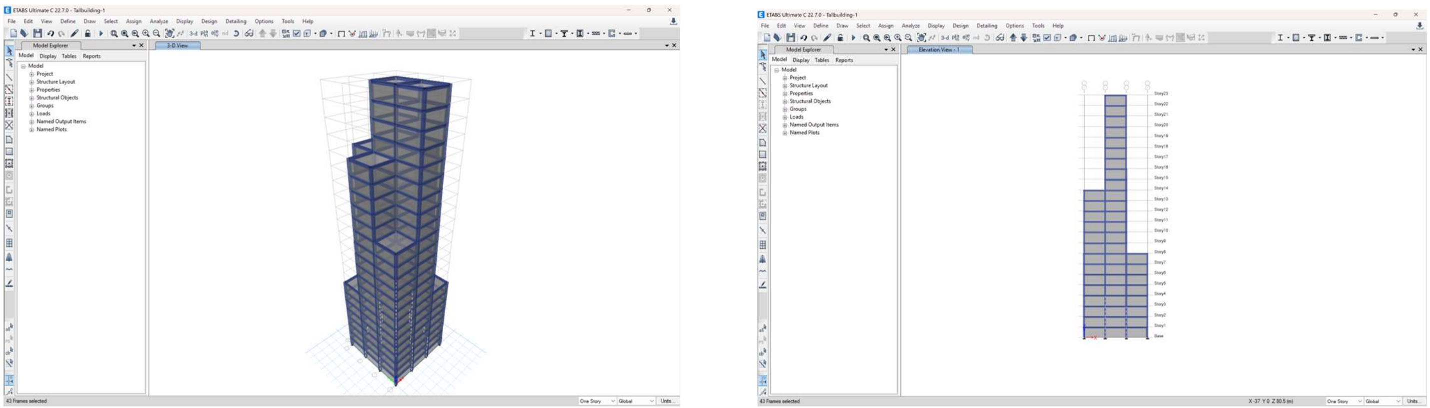

Figure 1: 3D Visualization and Elevation view of the building model.

2. Site Conditions & Seismic Parameters

The building site is Rupnagar, Punjab, which is classified under Seismic Zone IV as per IS 1893:2016. High seismicity necessitates a ductile design approach.

Seismic Zone Factor (Z)	0.24 (Zone IV)
Importance Factor (I)	1.2
Site Type	II (Medium Soil)
Response Reduction (R)	3.0

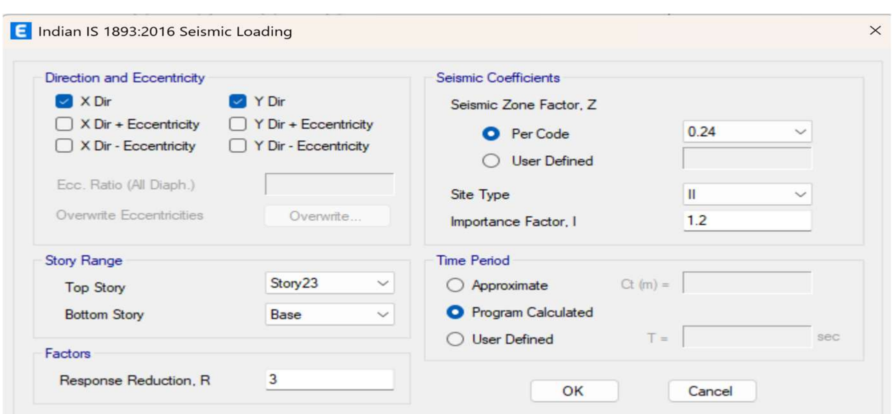

Figure 2: ETABS Input for Indian IS 1893:2016 Seismic Loading.

3. Load Combinations & Analysis

Standard load combinations were applied per IS 456 and IS 1893. The critical combination for checking lateral stability and displacement was 1.5(Dead) - 1.5(Seismic).

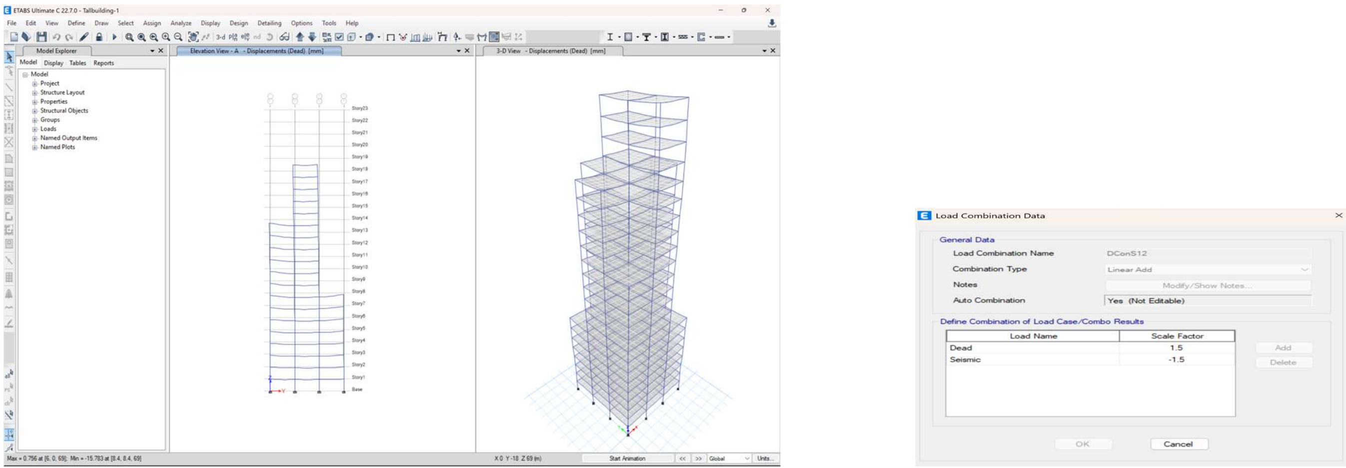

Figure 3: Definition of critical Load Combination (DCons12).

4. Maximum Story Displacement

A critical serviceability check for high-rise buildings is the story displacement and drift. The graph below shows the performance of the building under the critical lateral load combo.

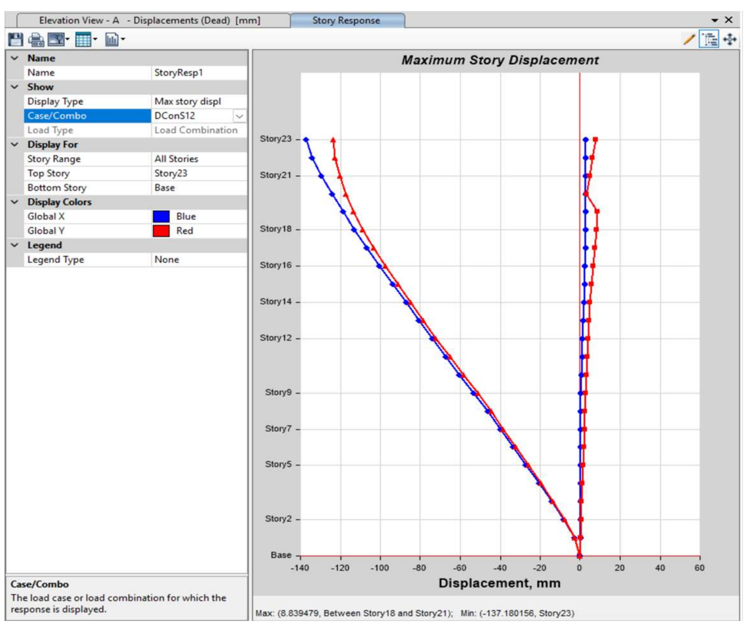

Figure 4: Story response plot for Maximum Story Displacement.

5. Concrete Design & Reinforcement

Design was carried out according to IS 456:2000. Longitudinal reinforcement was calculated for all members, ensuring that the heavy demands of Zone IV are met by the column sections.

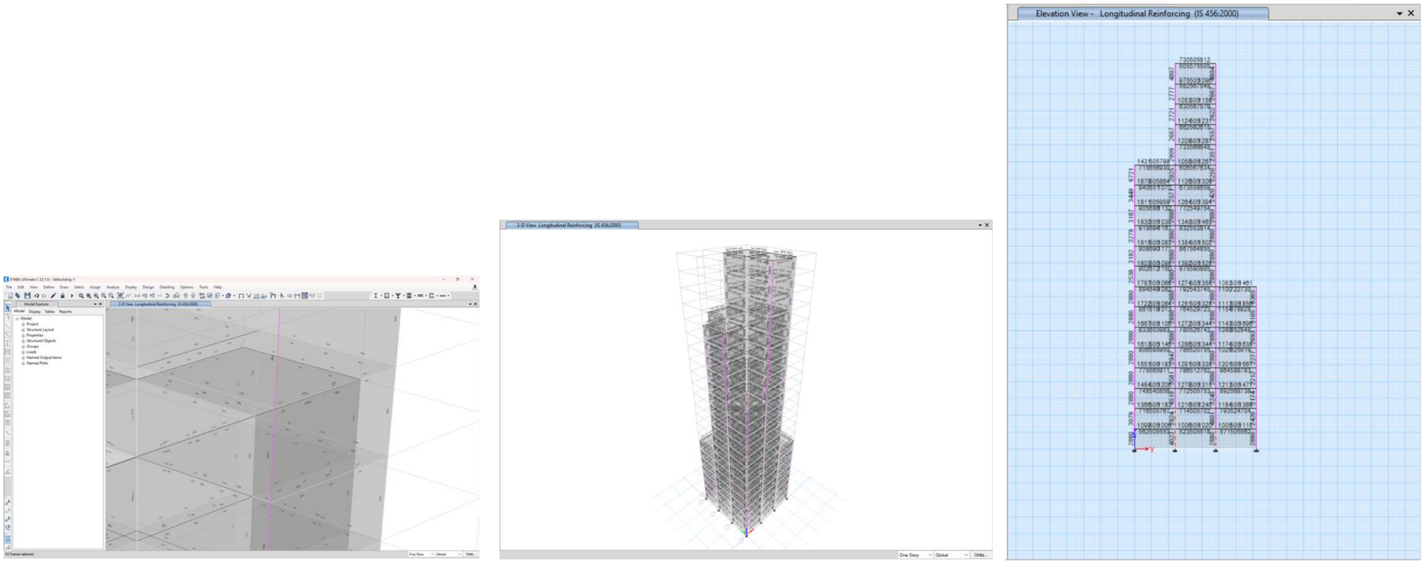

Figure 5: 3D reinforcement area mapping (IS 456:2000).

6. Conclusion

The main goal of this project was to bridge the gap between classroom theory and practical software application using ETABS. As a student, this case study provided significant learning opportunities:

- **Practical Code Application:** I learned how to interpret and apply **IS 1893:2016** parameters for a high-seismicity region (Zone IV), moving beyond basic textbook examples.
- **Software Proficiency:** Modeling a 23-story building with setbacks helped me understand how to manage complex geometries and interpret analysis results in a professional environment.
- **Safety vs. Design:** By checking story displacement and drift, I gained a better understanding of why serviceability limits are just as critical as strength requirements.
- **Engineering Judgment:** Seeing high reinforcement demands in the columns taught me about the real-world challenges of designing safe, constructible structures in earthquake-prone areas.

Overall, this project was a vital step in my development, helping me understand the responsibilities of a structural engineer and the importance of precision in seismic design.